

Q3.2 No Perfect Collinearity

The No Perfect Collinearity assumption requires that independent variables are not perfectly linearly related. We can check this with a correlation matrix and Variance Inflation Factors.

```
###  
## [1] "Correlation Matrix:"  
###  
  
###  
##           dreaming non_dreaming  
## dreaming      1.0000000    0.5142539  
## non_dreaming  0.5142539    1.0000000  
###  
  
###  
## [1] "VIF Scores:"  
###  
  
###  
##      dreaming non_dreaming  
##      1.35954      1.35954  
###
```

The correlation between dreaming and non_dreaming is -0.53, which is moderate. The VIF scores for both predictors are 1.39, well below the common threshold of 5 or 10. Though upon first glance, these variables may seem like perfect collinearity, this does not align with the data.

This model has no perfect collinearity due to this data.